Design of Sleeve-type Force Myography System Utilizing Thin Flexible Force Sensor using Weaved Optical Fiber

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***Abstract*— This paper introduces a novel sleeve-type force myography (FMG) system with a thin, flexible force sensor using weaved optical fibers for motion recognition. The compact sensor (15 mm width, 25 mm height, 2 mm thickness) exhibits high sensitivity and a wide force sensing range (>10N). Accuracy in force estimation exceeds 99.17%, with an 85 ms response time, ideal for real-time applications. The sensor design allows comfortable clothing embedding, highlighting its potential for prosthetics and virtual reality interactions. Proof of concept involves elbow flexion angle estimation, achieving 94.27% correlation coefficient. The proposed FMG sensor offers a promising solution for precise and intuitive human-robot interaction.**

I. Introduction

As the integration of robots extends into various aspects of our daily lives, human-robot interaction (HRI) assumes a pivotal role, especially in the fields of wearables, prosthetics, and virtual reality (VR) [1]-[3]. The challenges posed by current motion recognition methods underscore the need for innovative solutions. Traditional approaches, such as camera-based motion capture, often involve complex setups with multiple cameras, while glove-type sensors may constrain the placement of actuators, particularly in intricate VR scenarios requiring virtual haptic feedback [4]. Bio-signals like surface electromyography (sEMG) have been widely employed for motion recognition, detecting electrical stimuli from the skin. However, sEMG poses challenges such as the need for complex filtering processes and susceptibility to sweat and electromagnetic interference, limiting its usability [5]. In this context, force myography (FMG) emerges as a compelling alternative for motion recognition.

Among various FMG sensor types, optical fiber-based solutions stand out for their thin and lightweight nature, immunity to EMI and electrical noise, and high sensitivity [6]. This paper proposes a novel design of a thin and flexible force sensor utilizing weaved optical fiber for enhanced motion recognition, as shown in Fig. 1 (a). The proposed sensor, with a 2 mm thickness, offers high flexibility, making it easily embeddable in clothing or sleeves without causing discomfort. Notably, this sensor exhibits both high sensitivity and the ability to sense a wide range of forces (>10N), ensuring accurate and reliable performance. Evaluation results showcase its capability to estimate elbow flexion angles, affirming its potential as a versatile solution for intuitive and accurate motion recognition in diverse HRI applications.

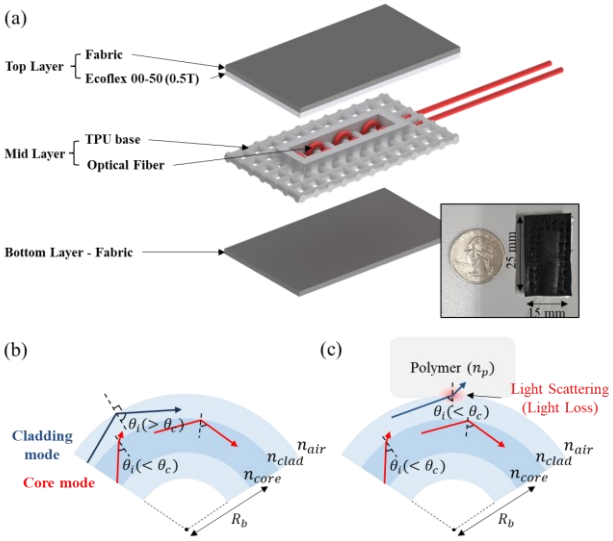


Figure 1. (a) Overall design of the proposed FMG sensor. The comparison of light loss when an optical fiber is bent (b) without and (c) with the contact with a polymer.

II. FMG Sensor Design using Optical Fiber

A. Working Principle of proposed sensor

Optical fibers employ total internal reflection at the core-cladding interface. Core-mode and cladding-mode represent light transmitted through the core and cladding, respectively. Optical fibers, with high sensitivity, find applications in sensors through light intensity modulation achieved by bending. When bent, the optical fiber allows some light to transition from core-mode to cladding-mode, impacting light loss. In situations with air exposure, only a small fraction of cladding-mode light refracts, causing minimal loss. However, when the cladding contacts a higher refractive index material like polymer ($n_p \gg n_{air}$), significant light loss and scattering occur as shown in Fig. 1 (b) and (c). For this reason, it is possible to create a highly sensitive force sensor that responds even to small touches. Achieving uniform bending methods, such as weaving, is explored. This study adopts weaving on a flexible base frame for its simplicity and larger contact area with the polymer.

B. Geometry and Fabrication Process

A 6-cycle weaving structure was employed using a 250 μm diameter plastic optical fiber (Eska, SH-1001.1) on a 3D printed base frame made of flexible thermoplastic

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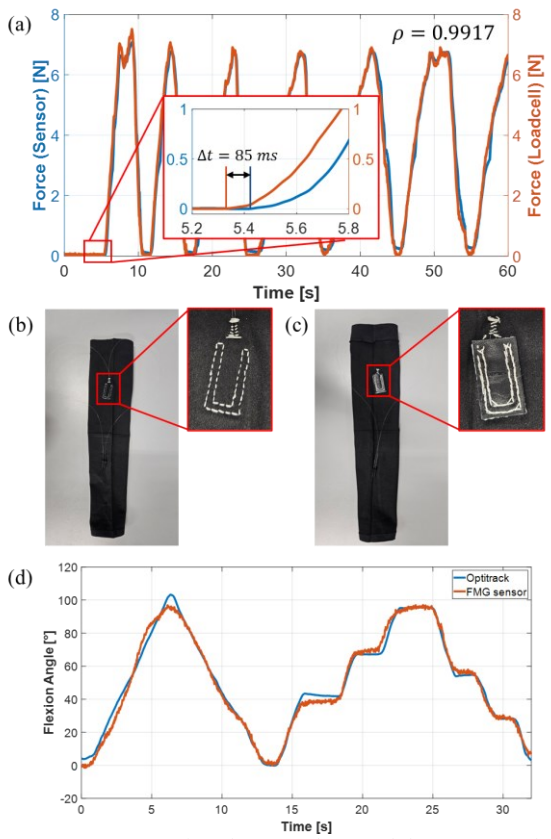


Figure 2. (a) Comparison between the actual force measured with a load cell and the force calibrated using the proposed sensor and its fast response time of 85 ms. (b) External and (c) internal of the FMG sensor embedded sleeve on the bicep muscle position. (d) User test result of flexion angle estimation using proposed FMG sensor.

polyurethane (TPU) with a 0.75mm bending radius. To maintain optical fiber alignment, the weaving direction reversed after 3 cycles. A wall structure around the optical fiber prevented significant light loss in case of direct polymer contact, with the wall thickness matching the optical fiber (250 μm). A polymer coating (Smooth-On, Ecoflex 00-50) of 0.5 mm thickness on a stretchable fabric enhanced mechanical dynamic response and served as a medium for optical fiber contact. The sensor, approximately 15 mm wide, 25 mm high, and 2 mm thick, offers a compact, thin, and highly flexible design, ideal for comfortable embedding in sleeves or similar applications. Its wearability ensures versatility without causing discomfort.

III. Performance Evaluation

A. Performance of the Proposed Sensor

The sensor response was examined by applying forces of approximately 10 N or higher using a load cell, and the results were subjected to polynomial curve fitting for calibration ($R^2 > 0.9$). To assess calibration performance, the force measured by the load cell was compared with the estimated force from the calibrated FMG sensor during multiple presses as shown in Fig. 2 (a). The correlation coefficient is approximately 0.9917, indicating a high level of similarity. And, the sensor response time was approximately 85 ms, meeting the requirement of not exceeding 300 ms for natural movements in prosthetics for amputees [7].

B. User Test – Elbow Flexion Angle Estimation

The sensor was embedded in the bicep position of the sleeve, as shown in Fig. 2 (b) and (c). A preliminary user test with one healthy male subject validated elbow flexion angle estimation using proposed FMG system. Using passive markers and a motion capture system (Optitrack, V120:trio), actual angles were compared with FMG estimates in real-time. The sensor measures normalized light loss when the bicep muscle presses on it, converting it to pressing force, and fur ther calibration and biomechanical modeling enable accurate flexion angle estimation during repetitive flexion and extension movements. The user test involved flexion and extension at 90-degree intervals initially, followed by 30-degree intervals. The experimental results, comparing actual arm angles with FMG estimates, demonstrated highly accurate flexion angle estimation (correlation coefficient ≈ 0.9427) as shown in Fig. 2 (d). The system effectively measured a wide range, including small angles, with sufficient precision.

IV. Conclusion

We proposed a thin and flexible force sensor using woven optical fiber for motion recognition. The sensor design leverages the principle of macro-bending loss and light scattering by touching polymers in optical fibers. The weaving structure of the optical fiber allows for a broad force sensing range (>10 N), and the sensor response time was measured at approximately 85 ms, ensuring rapid and accurate detection of pressing force. It demonstrated promising characteristics, with a thickness of 2 mm and high flexibility, making it suitable for embedding in clothes or sleeves without causing discomfort. The evaluation of the proposed concept involved using a single sensor to estimate elbow flexion angles, and the sensor successfully confirmed its capability in this regard. The results showed a high correlation coefficient of approximately 0.9427 between the actual arm angles and the estimated elbow flexion angles from the proposed sensor. The system exhibited not only high accuracy across a wide range of angles but also precision in measuring small angles.

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